Cell-type-specific compositional and transcriptional changes in mouse and human aging

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Abstract

We tested whether aging imposes one uniform transcriptional program across cell types. Two hypotheses made the idea testable. H2: the share of each cell type within an organ shifts in a conserved direction with age. H3: cells lose identity with age and transcribe more genes that do not belong to their program. We measured cell-type fractions, gene expression within specific cell types, and whole-profile distance from young references in mouse and human single-cell data, with exploratory arms in fly, worm, zebrafish, and naked mole rat. Fractions shifted only in specific cell types in specific organs, and the clearest human lung examples did not survive study-aware reanalysis as within-study age effects. A mouse-derived gene signature did not transport to either human cohort beyond expression-matched random signatures. Whole expression profiles drifted away from young references in mouse but showed only a small overall change in human, and the strongest human change was gene change in three interneuron subtypes that replicated across two cohorts, as part of broader age-associated expression change across cell types, without general drift of their whole profiles. Gene-level age effects shared direction across cell types more than a donor-level null allows in mouse. No statistically significant increase in off-program expression was detected in mouse lung at the tested depths, but confidence intervals were wide and biologically relevant increases were not excluded. Together, no uniform transcriptomic program was detected in the surveyed datasets. Upstream causes are not settled, and shared mechanisms with cell-type-specific consequences are not excluded.

Introduction

Two views of aging predict different transcriptomes in old tissue. In the first view, one upstream failure drives the rest, so old cells of every type should show the same broad pattern: weakened identity, transcription of genes that do not belong to their program, and movement away from the young state in a shared direction. In the second view, aging is a label for several time associated processes that co occur and interact, with different weights in different tissues and cell types. That view predicts heterogeneity instead: some cell types change and others do not, directions differ, and no single pattern appears everywhere. Recent work has claimed universal transcriptomic hallmarks of mammalian aging [11], which makes the distinction testable rather than rhetorical, and single-cell studies have already shown that transcriptional variability rises with age in immune cells [15] and that human aging signatures differ by cell type in pancreas [16].

There is no agreed operational definition of aging in general. Intuitively, aging is loss of function after passage of time: tissues work less well as time passes, repair weakens, and the risk of disease and death rises. That intuition is not yet an operational definition, because it does not say what to measure. This paper therefore states its working operational definition plainly. In this study, aging is measured as three things: (a) cell-type fractions, the share of each cell type within an organ; (b) gene expression within specific cell types; and (c) the distance of a whole expression profile from a young reference. That is the paper's working definition for the analyses reported here, not a final word on

aging itself. Any claim in this paper about aging is a claim about those three measurements, and any theory the paper constrains is constrained through them.

The word universal also needs an operational definition, because the headline claim depends on it. A uniform transcriptional program, as tested here, would have to satisfy four criteria: the same direction of change in every cell type tested, not only in a subset; overlapping gene sets driving the change, not only correlated directions; comparable magnitude after scaling by species lifespan; and presence in each species tested, not only in one. Failing any one criterion does not prove that no shared upstream process exists. It shows only that no uniform program was detected at the transcriptomic level under these criteria. A shared upstream process whose consequences differ by cell type would fail these criteria by design, and it is not excluded by this paper.

Bulk tissue measurements cannot separate these views, because a bulk change can reflect a shift in which cells are present rather than a change inside any cell. Single cell and single nucleus atlases allow composition and within cell expression to be measured separately, provided that sampling depth, detection differences between studies, reference choice, and the large number of cell type tests are handled as part of the inference rather than assumed away [1, 2].

This study, Agescape, was designed around those separations. It asks whether a mouse derived transcriptional signature transports to human cohorts; whether the share of each cell type within an organ changes with age; whether complete expression profiles move away from young references; whether cells express more genes characteristic of other cell types as they age; and whether the gene changes seen in human brain interneurons are specific to interneurons or part of broader change across cell types. A formal concordance test asks a narrower and answerable version of the universality question: do gene level age effects share direction across cell types within a tissue or cohort more than permuted age labels allow? Exploratory analyses in fruit fly, worm, zebrafish, and naked mole rat are reported as supplementary material. They do not drive the title or the main claims.

The scope is stated at the outset. Every measurement here is transcriptomic. Chromatin, DNA damage, transposon RNA, protein, cell function, and actual cell turnover were not measured. The results therefore test downstream transcriptomic predictions of mechanistic theories. They do not test the upstream mechanisms themselves, and no claim in this paper depends on pretending otherwise.

Materials and Methods

Datasets and inferential units

The mouse atlas was Tabula Muris Senis droplet data distributed through CZ CELLxGENE, with 23 mouse identifiers and 238,186 cleaned cells [1]. Public human brain, kidney, and lung data came from the CZ CELLxGENE Census release of 2025-11-08, with 621 control donors for signature scoring and 479 cohort specific donors with adequate young reference coverage for expression vectors [2]. The human lifespan cortex arm was the PsychAD dorsolateral prefrontal cortex lifespan atlas (Synapse syn52396927, with the pseudobulk resource syn62064518), with 284 control donors and 27 cell types [9]. The integrated human lung cohort came from the lung cell atlas of Sikkema and colleagues, with 119 donors and 44 cell types [10]. Exploratory arms were the naked mole rat skin data of Savina and colleagues (GEO GSE152886, five animals after exclusion of sample 2V) [7, 8], the Aging Fly Cell Atlas (GEO GSE218661, 36 age by sex pools in head and body) [3], the Gao worm atlas (GEO GSE229022, 12 wild type pools) [4], the Roux worm atlas (GEO GSE208154, descriptive only) [5], and zebrafish testis (GEO GSE275361, ten males for broad compartments and six fish for fine grained analyses) [6]. Only primary, normal disease human cells were included, and public human analyses used donors aged at least 20 years. Donors, animals, fish, or pools were the inferential units, not individual cells. Depth is the total number of RNA molecules captured per cell, summarized as mean log UMI.

Cohort	Donors or animals	Ages	Sexes	Tissues	Platform	Cells
Mouse, Tabula Muris Senis droplet atlas	23 mouse identifiers	Young reference 1 to 3 months; full age range not reported here	Male and female	12 tissues, 62 tissue by cell type units	Droplet single-cell RNA sequencing	238,186 cleaned cells
Human, CZ CELLxGENE Census public cohorts	621 control donors for signature scoring; 479 donors for expression vectors	At least 20 years	Male and female	Brain, kidney, lung	Not reported	Not reported
Human, PsychAD lifespan cortex atlas	284 control donors	Not reported here	Male and female	Dorsolateral prefrontal cortex, 27 cell types	Not reported	Not reported
Human, integrated lung cohort	119 donors	20 to 81 years	Male and female	Lung, 44 cell types	10x variants and Seq-Well, known for 66 of 119 donors	Not reported
Naked mole rat, skin	5 animals	Group level ages only	Not reported	Skin only	Not reported	Not reported
Fruit fly, Aging Fly Cell Atlas	36 age by sex pools	Not reported here	Male and female pools	Head and body	Not reported	Not reported
Worm, Gao atlas	12 wild type pools	Not reported here	Not reported	Whole animal	Not reported	Not reported
Zebrafish, testis	10 males for broad compartments; 6 fish for fine grained analyses	Not reported here	Male only	Testis	Not reported	Not reported

Table 1. Cohorts analyzed. Cells marked not reported were not available in verified form in the result files and were not estimated.

Mouse processing and differential expression

Mouse cells were filtered using the prespecified atlas quality control rules: 500 to 5,000 detected genes, at least 1,000 UMIs, and doublet removal. Mitochondrial filtering was not applicable because the released droplet object lacked mitochondrial transcripts. Counts were normalized to 10,000 per cell and transformed with log1p, using SCANPY [21]. For differential expression, donor by cell type pseudobulk counts were aggregated and log transformed after counts per million scaling. Age effects were estimated within each cell type with sex included when identifiable. P values used t based regression inference and Benjamini-Hochberg correction within the stated family [23]. Per-gene calls were checked for method sensitivity: the primary calls used DESeq2, and moderated weighted least squares was run as a sensitivity analysis. The two methods give very different call counts, as reported in the Results, so no per-gene conclusion in this paper rests on one method alone.

Signature derivation and transport

P1_Clean contained 139 mouse genes, 89 up and 50 down. P0_Raw257 contained 257 genes and was retained as a sensitivity score. P1 used uniform Benjamini-Hochberg differential expression candidates from adequately covered mouse units and removed predicted or pseudogene labels, non protein coding

biotypes, ribosomal and mitochondrial genes, and specified ambient RNA cases. Sex dimorphic and immediate early filters belonged to P2 and P3, not P1. One to one mouse to human orthologs came from the Ensembl derived mapping, giving 118 mapped P1 genes. The score was mean expression of up regulated genes minus mean expression of down regulated genes, with missing genes omitted rather than imputed. Public cohort regressions included age and sex when variable. Primary PsychAD models included age, sex, brain bank, postmortem interval, and three technical principal components. One Benjamini-Hochberg family covered 225 public cohort cell type tests and 27 primary PsychAD tests, for 252 tests. The matched null analysis in PsychAD generated 200 random signatures per cell type, each matching the detected P1 up and down counts and expression decile distribution, excluding P1 genes from the candidate pool, and using the same score and fully adjusted regression. The corresponding matched null analysis in MSSM generated 200 random signatures per cell type on the same design, matching the detected P1 up and down counts and expression decile distribution, excluding P1 genes from the candidate pool, and using the same score with the age-plus-sex regression of the primary MSSM analysis. The Results state the PsychAD and MSSM null outcomes together.

Human interneuron differential expression and replication

Human brain differential expression used donor by cell type pseudobulk counts from the Census release, restricted to primary data from donors with normal disease status, at least 20 cells per donor by cell type, and ages 20 to 120 years. Counts were converted to log counts per 10,000. Age effects were estimated with precision-weighted linear models (a limma-voom style variance model, implemented without the limma package): per-gene residual variance was smoothed against mean expression to give precision weights, and standard errors were moderated by empirical Bayes shrinkage of the residual variances toward a fitted prior. Models included age, sex when variable, and donor mean log library size as a depth covariate. Benjamini-Hochberg correction was applied within each cohort by cell type over the tested genes. Replication required passing false discovery rate correction in both cohorts in the same direction, and was tested against a hypergeometric null and a block permutation null that permutes HBCC significance and sign labels within coexpression blocks (30 blocks from top principal component loadings of the pseudobulk data), preserving coarse coexpression structure. Leave-one-donor-out refits dropped each donor once and refit the same model. The same models were run across all 39 tested MSSM and HBCC cell types as a specificity check.

Cell type composition

A cell fraction is the share of one specific cell type among recovered cells in one organ, modeled as a centered log ratio slope. A fraction is never a tally across different denominators, never an absolute cell count, and a change in a fraction is not by itself evidence of fibrosis or of organ damage.

For donor i and cell type j, counts were transformed with pseudocount a and the centered log ratio: CLR_ij = log((count_ij + a) / sum_j(count_ij + a)) minus the mean over j of the same logged shares. The primary pseudocount was 0.5. Primary regressions included age and sex when identifiable. Sensitivity models added log total recovered cells and evaluated pseudocounts 0.1, 0.5, and 1.0, and a global Benjamini-Hochberg recheck applied one correction across the core mouse, human, and naked mole rat tests. Propeller provided a donor level count model check: proportions were transformed as logit((count + 0.5) / rowSum(count + 0.5)), followed by limma empirical Bayes regression with age, log total cells, sex when variable, and platform for Gao worm [19, 20]. Recovered composition was not interpreted as absolute tissue cell number. Study-aware models for human lung added study fixed effects, per-study slopes, and fixed and random effects meta-combination across studies, as reported in the Results.

Whole expression vector displacement

An expression vector is a complete within-cell expression proportion vector. No UMI downsampling was used for the direct vector analysis. Mitochondrial, ribosomal, and sex linked marker genes were excluded using species appropriate symbols. Each cell raw count vector was converted to within cell proportions p, and a young reference probability vector q was constructed from pooled counts of the same annotated cell type, with young donors or animals excluded from their own reference. Hellinger distance was H(p,q) = sqrt(1 minus sum over g of sqrt(p_g q_g)). Each evaluated donor by cell type unit required at least 20 cells and at least 50 remaining young reference cells. Mouse young references used animals aged 1 or 3 months, and human references used donors aged 20 to 40 years. Continuous age models included sex and, in specified arms, mean log UMI. Species summaries used unit fixed intercepts and donor clustered standard errors. Effects were scaled to 10 percent of species maximum lifespan using AnAge values of 4 years for mouse, 122.5 years for human, and 31 years for naked mole rat [22]. The mouse leave one cell type out analysis removed each cell type across all tissues and refit the global model. Depth sensitivity arms fixed sequencing depth at 500, 1,500, and 3,000 UMIs per cell and at full depth, and a pseudobulk Hellinger arm aggregated cells before measuring distance. Young versus young controls compared young animals against a split young reference to test for spurious displacement. An 18 month reference arm tested whether drift continues past maturity.

Gene effect concordance across cell types

Donor pseudobulk counts were aggregated per cell type, normalized to counts per 10,000, and log1p transformed. Each cell type yielded a vector of gene level age effects from models including age, sex when variable, mean log UMI, and platform for Gao worm. Pairwise Pearson and Spearman correlations were computed between cell type effect vectors within a tissue or cohort, and the primary summary was mean pairwise Pearson correlation. The primary null is a joint donor-level permutation: one age assignment per donor, shared across all of that donor's cell types, permuted within sex strata, with 1,000 permutations. This null preserves the correlation that a single donor contributes to many cell types. Per-cell-type permutation, which permutes age labels separately within each cell type, is shown only as a superseded comparison, because it breaks donor sharing and understates the null. Human, fly, worm, and zebrafish concordance values were not redone under the joint null, because the underlying data were not available locally; their values are reported as reference only, with their p-values treated as invalid.

Off program expression, entropy, and depth sensitivity

Off program expression means expression of genes that are uncommon in young cells of the target type but characteristic of another cell type. That outcome is never called entropy in this paper. A separate Shannon entropy measure is reported under its own name where it was computed.

Within cell Shannon entropy was minus the sum over g of p_g log(p_g), after the stated gene exclusions. Off program genes were uncommon in young target cells but characteristic of another lung cell type: detection in at most 5 percent of young target cells and at least 20 percent of another cell type, expression of at least 20 counts per 10,000, and at least fivefold enrichment over the target. Young references excluded the evaluated young animal. Animal means were regressed on age and sex with Benjamini-Hochberg correction across cell types. The original analysis downsampled all cells without replacement to 1,500 UMIs. Depth sensitivity analyses evaluated 500, 1,500, and 3,000 UMIs and full depth, with no downsampling in the full depth arm, the original 1,500 UMI inclusion threshold retained, young reference marker definitions fixed across depths, and off program fractions computed against actual total UMIs. The library size arm added each animal mean log actual pre downsampling depth. Global models used cell type intercepts and animal clustered t based inference.

Human nonlinear age models and annotation stability

Nonlinear age effects used a cubic B spline with three degrees of freedom, fixed before examining results. F tests compared nonlinear and linear models on the same donor by cell type units, with Benjamini-Hochberg correction within cohort separately for linear and nonlinear comparisons. For annotation stability, excluded and unassigned cells were tabulated by age and tissue or cohort, and Spearman tests assessed age associations of excluded and unassigned fractions. A fixed canonical marker panel was evaluated by donor by cell type log counts per 10,000 expression with age, sex, and depth covariates, requiring at least six donors and 20 cells per donor by cell type unit, with Benjamini-Hochberg correction across targeted marker tests. Marker change was not equated with identity loss.

Statistical families and multiplicity

Each analysis family was defined before correction, and Benjamini-Hochberg correction [23] was applied within the stated family. The families are: mouse differential expression within each tissue by cell type unit; signature transport, one family of 252 tests across the public cohorts and PsychAD; composition, first within each organ or cohort and then one global family across the core mouse, human, and naked mole rat tests; whole expression vectors, within each species and cohort; concordance, one joint permutation test per species and scope; off-program expression and entropy, within the mouse lung depth sweep; interneuron gene replication, within each subtype and cohort; and nonlinear age models, within cohort separately for linear and nonlinear comparisons. Non-significance was not treated as equivalence. No equivalence claim is made in this paper.

Reproducibility and reporting

Aggregate result tables were retained for every reported number. Controlled PsychAD donor level values and identifiers were not exported. Exploratory, secondary, and primary families were labeled explicitly.

Results

Cell fractions change locally, in specific types and organs

Fraction changes (the share of one specific cell type among recovered cells in one organ, modeled as a centered log ratio slope) were found in some organs and datasets and not in others, and always in specific cell types. Composition tests were first corrected within each organ or cohort; a wider correction across the core mouse, human, and naked mole rat tests was then applied. Under that wider family, mouse retained at least one significant fraction change in seven organs, down from eight, and tongue no longer qualified. Human data retained at least one change in two cohort datasets, down from three, and Kidney Living Donor no longer qualified. MSSM retained three changes and integrated lung retained two under the global correction (Figure 1).

Three sensitivity analyses support reading these as fraction changes among recovered cells, and nothing more. Models adjusted for log total recovered cells gave, within their own strata, 15 significant mouse tests across six organs and ten human tests across three cohort datasets. Pseudocounts of 0.1, 0.5, and 1.0 gave strongly correlated effects: the correlation between the 0.1 and 1.0 arms was 0.9569 in mouse and 0.9434 in human, and the share of tests with the same sign in all three arms was 0.9412 in mouse and 0.9313 in human. A propeller logit plus limma check agreed closely with the centered log ratio models, with Spearman rho of 0.9738 in mouse and 0.9313 in human, and with ten mouse and nine human results passing false discovery rate correction in both models. These models change some individual significance calls. They do not turn recovered cell counts into direct measurements

of absolute tissue composition, and no fraction result in this paper demonstrates fibrosis or organ destruction.

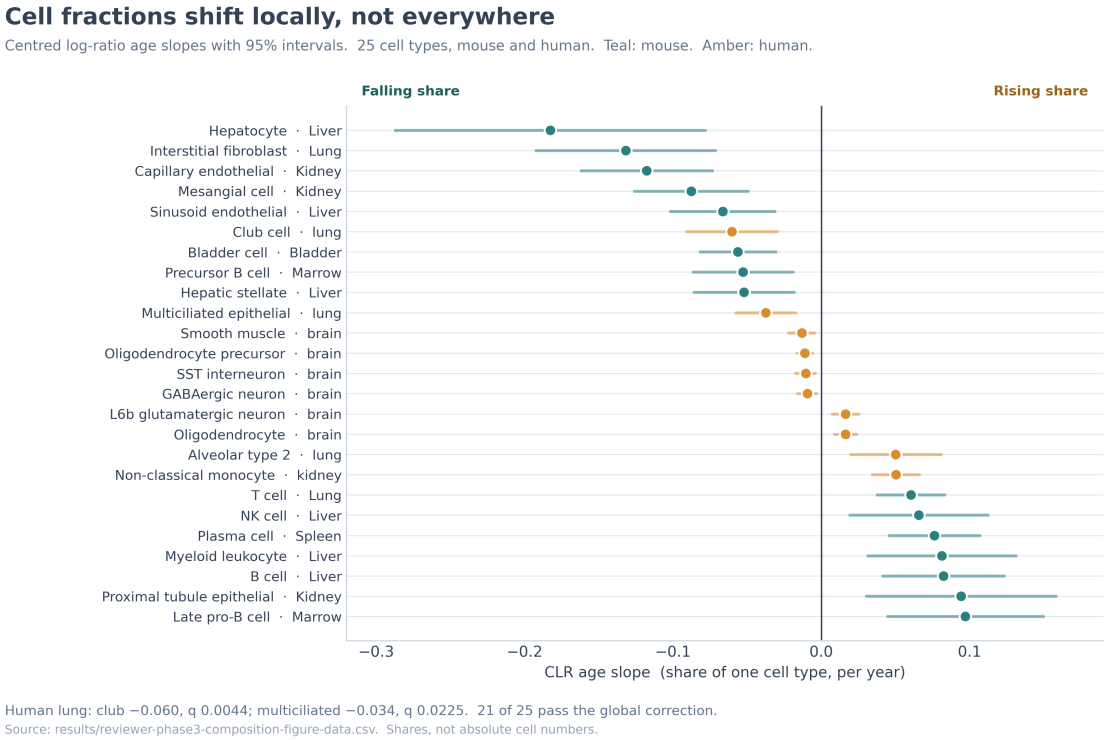


Figure 1: Figure 1. Cell fraction age slopes with 95 percent confidence intervals. The 25 results shown are the mouse and human results passing false discovery rate correction within their own strata at pseudocount 0.5, sorted by slope; 21 of the 25 also pass the wider global correction.

The human lung example, previously the clearest named case, does not survive study-aware reanalysis as a within-study age effect. In the pooled integrated lung cohort of 119 donors and 44 cell types, club cells decline with age (slope minus 0.060227, q = 0.004423) and multiciliated epithelial cells decline with age (slope minus 0.034154, q = 0.022513). Once study is modeled, the picture changes. With study fixed effects, the club slope falls to minus 0.0154 (p = 0.103) and the multiciliated slope falls to minus 0.0056 (p = 0.467). Within individual studies, no study replicates the club decline at p less than 0.05 in any of 10 estimable studies, although 8 of 10 slopes are negative. Meta-combination across studies gives a club fixed effect of minus 0.0159 (95 percent confidence interval minus 0.0296 to minus 0.0022, p = 0.023) and a random effects estimate of minus 0.0158 (p = 0.036), and a multiciliated fixed effect of plus 0.0098 (p = 0.095) with a random effects estimate of plus 0.0075 (p = 0.487), so the multiciliated sign flips and the effect is dead. The pooled club and multiciliated findings are therefore downgraded to between-study associations, not presented as age effects.

Three further facts explain why the pooled lung result is fragile. First, the recovered club fraction spans 0.00009 to 0.3075 across studies: young studies sampled airway and nose, where club cells are common, while old studies sampled parenchyma, where they are rare, so study and age are entangled with sampling site. Second, the excluded-cell fraction rises with age, from 16.1 percent at ages 20 to 39 to 34.6 percent at ages 60 to 81, so older donors contribute systematically fewer annotated cells. Third, the earlier club-cell identity test was invalid: a B-cell marker rule fired before the club-cell rule, so club cells were tested against the B-cell marker CD79A, and club identity loss is untested, not excluded. An independent lung cohort of 17 donors also fails to replicate the finding, with opposite-sign slopes in its nearest available airway analogues. The lung fraction results are reported here as associations that

track study and sampling differences, not as established age effects.

A mouse derived signature does not transport beyond matched random signatures in either human cohort

The primary signature, P1, contains 139 mouse derived genes, 89 up regulated and 50 down regulated with age, after technical and biotype filtering. The 257 gene P0 signature is a sensitivity analysis only. P1 was derived from the mouse differential expression analysis before the reported human transport evaluations, and no human result was used to choose its membership. One to one mouse to human mapping yields 118 genes, 76 up and 42 down, with missing genes omitted rather than imputed.

The signature derivation itself is method sensitive, and that sensitivity bounds everything that follows. Moderated weighted least squares calls 151 genes, DESeq2 calls 312, and unmoderated weighted least squares calls 1, across the same 17 mouse units. Eight of the 17 units give zero moderated calls. In leave-one-donor-out refits, 310 of 765 P1 gene by unit instances change sign, or 40.5 percent. Per-gene calls therefore depend heavily on the differential expression method and are unstable to single-donor removal, and the signature should be read as one method-dependent gene set, not as a fixed biological program.

Across 252 primary human cell type tests under one shared correction, only three survived: MSSM layer 2/3 neurons, MSSM oligodendrocytes, and PsychAD EN_L3_5_IT_3 neurons. The slopes are minus 0.001303 per year for MSSM layer 2/3 neurons ($q = 0.0208$), plus 0.001152 per year for MSSM oligodendrocytes ($q = 0.0346$), and minus 0.001383 per year for PsychAD EN_L3_5_IT_3 neurons ($q = 0.0208$). Re correction of the same 252 p values without outcome based deletion left the three survivors and their q values unchanged. Whole cortex transport was not detected. No oligodendrocyte precursor cell or microglial result survived the primary correction, so those results remain candidates, not established findings.

Two qualifications now bound this result. First, PsychAD is not independent replication of the public cohorts. Exact identifier matching established that 103 of 214 public MSSM donors and 131 of 134 public HBCC donors also appear in the corresponding PsychAD brain bank arms, with concordant age and sex. The MSSM to HBCC interneuron comparison below uses different source banks and is not reclassified by this overlap. Second, matched random signature analyses in both cohorts fail the three survivors. In PsychAD, 200 expression decile matched random signatures per cell type, using the same score and the fully adjusted model, leave no cell type surviving Benjamini-Hochberg correction; the smallest q value was 0.1007. In MSSM, 200 expression decile matched random signatures per cell type, using the same score and the age-plus-sex model, also leave no cell type surviving correction across the 21 MSSM types: the layer 2/3 neuron association gives an empirical p value of 0.0249 and q of 0.1045, and the oligodendrocyte association gives an empirical p value of 0.0050 and q of 0.0522, just above 0.05. Endothelial cells share that smallest empirical p value of 0.0050 with the same q of 0.0522, but were not a global survivor. No cell type survives both the global correction and the matched random signature null in either cohort. The null recomputation reproduced the primary MSSM slopes to within 0.000083 per year, so the comparison is like with like. The reported P1 associations should therefore be treated as not distinguishable from expression matched random signatures under this test, and the signature transport claim is dropped. The PsychAD matched null was not re-run in this revision: PsychAD is absent from the Census release used here, and its pseudobulk resource requires Synapse authentication, so its earlier outcome stands as reported.

Mouse whole profile displacement is distributed and survives depth adjustment, and the human overall displacement is small and imprecise

Mouse showed a robust global vector signal (an expression vector is a complete within-cell expression proportion vector, compared by Hellinger distance to a cross-fitted young reference from the same

cell type, with no per-cell downsampling). The recomputed unadjusted effect is plus 0.1027 standard deviations per 10 percent of maximum lifespan, matching the published plus 0.1028 ($p = 0.00000398$). Centroid drift survives fixed sequencing depths: at 500 UMIs the linear depth-adjusted effect is plus 0.0953 ($p = 0.0139$), at 1,500 UMIs plus 0.1278 ($p = 0.0016$), at 3,000 UMIs plus 0.1490 ($p = 0.00039$), and at full depth plus 0.0700 ($p = 0.000015$). Dispersion, the spread of cells around their own type center, died at every fixed depth: minus 0.0017 at 500 UMIs ($p = 0.96$), plus 0.0145 at 1,500 UMIs ($p = 0.68$), and plus 0.0288 at 3,000 UMIs ($p = 0.41$). The mouse signal is therefore displacement of cell type centers away from young references, not increased scatter within cell types. Pseudobulk distance shows the same gradient: plus 0.1529 at full depth ($p = 0.0000024$), plus 0.0926 at 1,500 UMIs ($p = 0.00019$), and plus 0.0480 at 500 UMIs ($p = 0.073$, not significant). Young versus young controls pass, with permutation p values of 0.335 and 0.298, so the method does not manufacture displacement where no age difference exists.

A decomposition shows that this signal is not one cell type in disguise. Leave one cell type out global effects ranged from plus 0.0970 to plus 0.1119 around the full estimate of plus 0.1028, and the largest absolute change from removing any one cell type was 0.00912 standard deviations. The largest reductions followed removal of mesenchymal stem cells, podocytes, fenestrated cells, lung fibroblasts, or endothelial cells. No single cell type removes the global association (Figure 2). The earlier exact sign test, 45 of 62 units positive with $p = 0.000497$, is dropped: it treated 62 correlated unit signs as independent trials, while the same 23 mice contribute to many units. In its place, a donor-cluster bootstrap resampling the 23 mice gives a mean fraction of positive units of 0.676 (95 percent prediction interval 0.561 to 0.792), with probability 0.0065 of a fraction at or below 0.5. Leave-one-donor-out sign counts range from 35 to 47 positive units, and leave-one-tissue-out counts range from 34 to 45, with the global effect ranging from plus 0.0903 to plus 0.1088 across leave-one-donor-out refits. The distributed pattern survives, but the evidence is weaker than the dropped sign test implied.

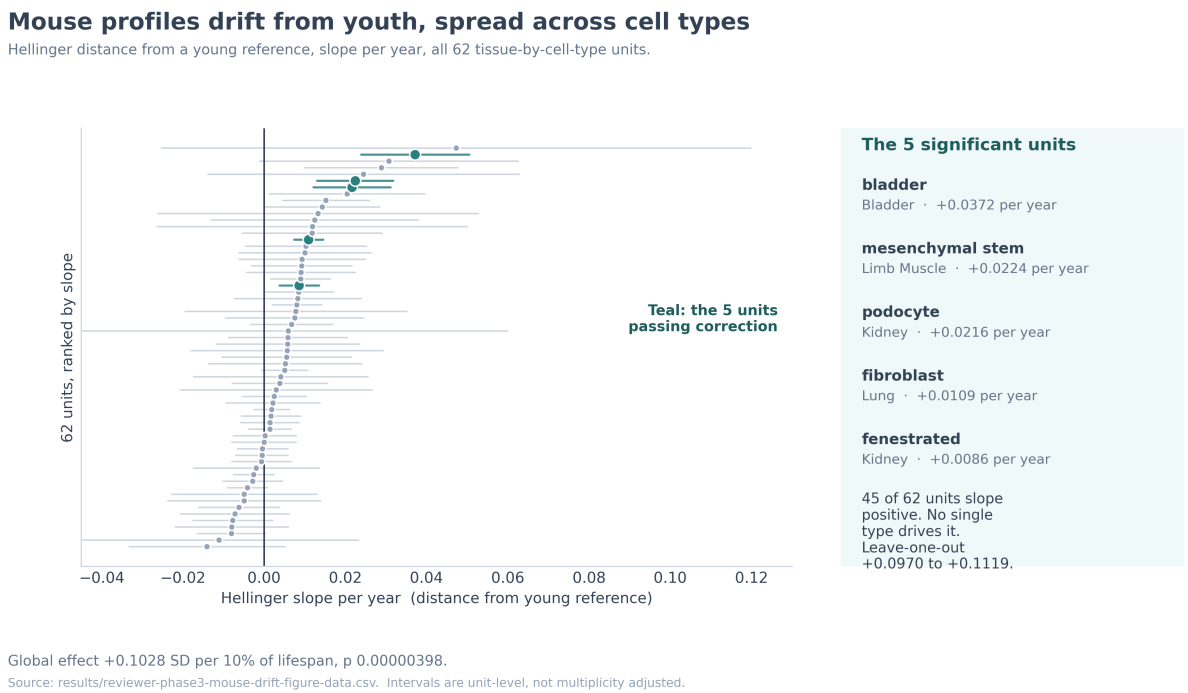


Figure 2: Figure 2. Mouse vector drift by cell type. Each point is a donor level age slope in mean Hellinger distance from a cross fitted young expression reference for one mouse tissue by cell type unit, ranked by slope; all 62 units are shown. Bars are 95 percent confidence intervals.

The mature reference arm changes where the drift sits in the lifespan. With an 18 month reference,

the effect dies after depth adjustment: the linear effect is plus 0.0072 (p = 0.67) and pseudobulk effects are minus 0.010 to minus 0.018. The drift is therefore concentrated in the young to mature transition, not in continued drift past 18 months, and that is stated plainly here.

Per-organ decomposition is uneven. Kidney is robust at all depths. Lung is significant at full depth (plus 0.0850, p = 0.0065) and at 3,000 UMIs (plus 0.155, p = 0.030). Tongue is plus 0.149 (p = 0.0046) and bladder is plus 0.131 (p = 0.048) at full depth. Spleen is significant only at 1,500 and 3,000 UMIs. Marrow is dead at every depth, a depth artifact in the original analysis. Heart and aorta is negative and never significant at cell level. The global mouse result is therefore carried by some organs and absent in others.

The human overall vector effect was plus 0.0365 standard deviations (p = 0.2218), with a 95 percent confidence interval of minus 0.0221 to plus 0.0951. That interval cannot distinguish no effect from a mouse-sized effect: the mouse point estimate of plus 0.1028 sits just 0.0077 standard deviations above the human interval upper bound. No equivalence claim is made. The mouse minus human interaction is minus 0.0663 unadjusted (95 percent confidence interval minus 0.1330 to plus 0.0004, p = 0.0515, bootstrap p = 0.072) and minus 0.0447 after depth adjustment (95 percent confidence interval minus 0.0760 to minus 0.0135, p = 0.0051, bootstrap p = 0.030). On a range scale the interaction is not significant (unadjusted p = 0.219, adjusted p = 0.057). Human vector effects are never robust to the depth model: linear plus 0.0231 (p = 0.050), inverse square root plus 0.0362 (p = 0.013), spline plus 0.0138 (p = 0.20). The honest report is the interval, not a verdict of absence.

Gene level age effects share direction across cell types, without establishing a universal program

The concordance test compares gene level age effect vectors between cell types within a tissue or cohort. The earlier per-cell-type permutation null is superseded: it permuted age labels separately within each cell type, breaking the sharing of one donor across many cell types and understating the null. The primary null is now a joint donor-level permutation, one age assignment per mouse shared across all its cell types, permuted within sex, with 1,000 permutations. In mouse, mean pairwise Pearson concordance within tissue was plus 0.1868, against a joint null mean of plus 0.1378 (standard deviation 0.0251, 95 percent interval plus 0.0973 to plus 0.1952), with p = 0.0430, across 78 units, 22 mice, and 11,019 genes. Cross-tissue concordance was plus 0.0363 against a null of plus 0.0234 (p = 0.155), which is not significant, so shared direction is a within-tissue phenomenon, not a cross-tissue one. The noise ceiling is 0.3339 and split-half reliability is 0.3786, so the observed concordance is about half of what perfect measurement would allow. Only two per-tissue strata survive the joint null: Heart and Aorta (p = 0.0020) and Mammary Gland (p = 0.0150). In naked mole rat, the null was already donor-level, and the observed plus 0.6146 against a null of plus 0.6423 (p = 0.70) is null. Human, fly, worm, and zebrafish were not redone under the joint null, because the underlying data were not available locally; their values (human plus 0.1981, fly plus 0.1936, worm plus 0.3537, zebrafish plus 0.3393) are reference only, and their p-values are invalid and are not reported as evidence (Figure 3).

This result cuts in a specific direction and no further. Shared direction of gene level effects is not evidence that every gene changes, that whole vector distance increases everywhere, that transcriptional disorder increases, or that the same genes respond across species. A prior report of a conserved AP-1 immune aging signature shows that shared direction across cell types has been seen before [12]; it does not show a universal program.

Human interneuron gene changes replicate across two cohorts, as part of broader age-associated change

The durable human brain result is replicated gene change in three interneuron subtypes. Donor pseudobulk models with depth adjustment and moderated standard errors (see Materials and Methods)

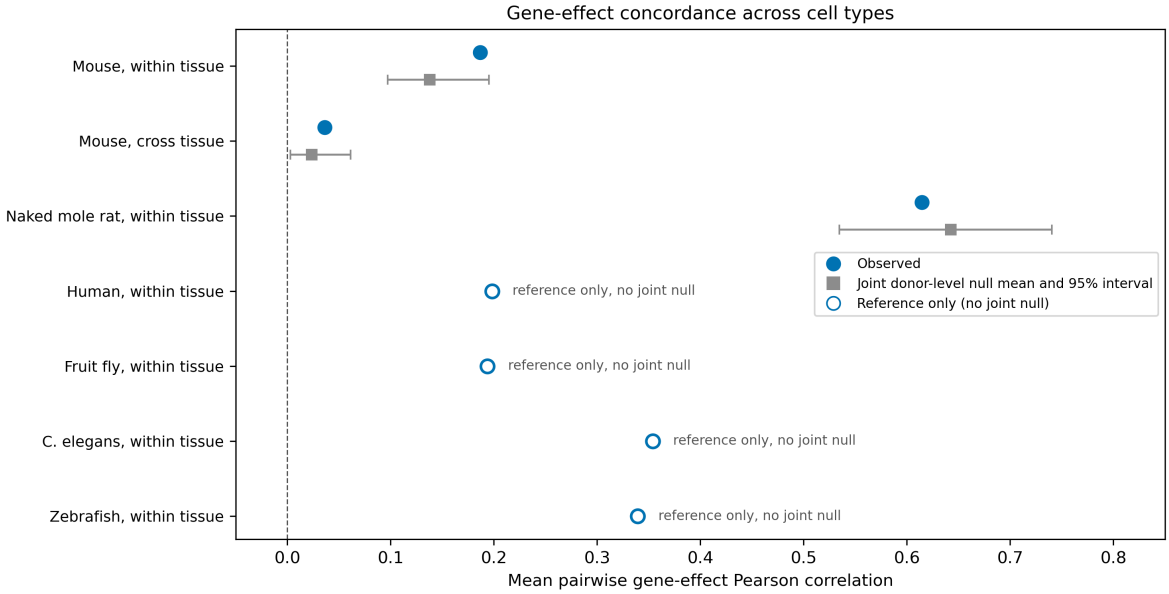


Figure 3: Figure 3. Gene effect concordance. Teal circles show observed mean pairwise Pearson correlations between cell type gene level age effect vectors; grey squares show the mean and 95 percent interval of the joint donor-level age permutation null. Rows without a null are reference only: the joint null was not computed for those species.

give, at false discovery rate below 0.05, 219 genes in VIP interneurons, 145 in parvalbumin interneurons, 385 in somatostatin interneurons, and 72 in the generic GABAergic neuron type in MSSM, and 213, 220, 583, and 40 respectively in HBCC. Moderation barely moves the counts: the unmoderated counts are 218, 152, 400, and 75 in MSSM and 224, 232, 604, and 45 in HBCC. Of the genes tested in both cohorts, 77 VIP genes, 65 parvalbumin genes, and 239 somatostatin genes pass false discovery rate in both cohorts in the same direction, 381 genes in total (on the shared tested sets, the per-cohort false discovery rate counts are 203 and 205 for VIP, 137 and 216 for parvalbumin, and 362 and 566 for somatostatin), against hypergeometric expectations of 22.5, 16.5, and 113.3 genes (p = 1.0e-27, 1.8e-27, and 1.8e-53; block permutation p = 0.001 for each subtype). The generic GABAergic neuron type does not replicate at all: 0 genes pass in both cohorts, against 1.5 expected (p = 1.0). MSSM VIP counts use 197 donors in this fetch, one fewer than the 198 in the primary signature analysis, because Donor_858, aged 19, is excluded by this pipeline's 20 to 120 year age filter.

Effect sizes, not only overlap counts, agree between the cohorts. Across all genes tested in both cohorts, not only the selected ones, the MSSM to HBCC effect size correlations are Pearson r = 0.722 and Spearman rho = 0.642 for VIP, r = 0.684 and rho = 0.649 for parvalbumin, and r = 0.761 and rho = 0.694 for somatostatin, with 74.6, 75.5, and 76.5 percent of genes changing in the same direction. Among the selected MSSM false discovery rate genes the correlations are higher (r = 0.855, 0.849, and 0.890), as expected for a selected set, and are reported here only as a secondary check. Same direction concordance among MSSM false discovery rate genes is 191 of 203 VIP genes, 135 of 137 parvalbumin genes, and 350 of 362 somatostatin genes. The three subtype sets also overlap each other above chance: 32 VIP and parvalbumin genes, 67 VIP and somatostatin genes, and 53 parvalbumin and somatostatin genes, with 17 genes shared by all three subtypes (Figure 4).

Leave-one-donor-out refits support stability of the replicated genes: across 782 gene-by-cohort refits of the replicated gene set, no slope changed sign, the median largest single-donor shift was 0.48 standard errors, only three cases shifted by more than one standard error, and only four cases lost nominal significance in any refit. Replication is real for the three subtypes but is not exclusive to interneurons.

The same models run across all 39 tested MSSM and HBCC cell types find extensive age-associated expression change elsewhere: in MSSM, 814 genes in microglia, 604 in oligodendrocytes, 562 in layer 2/3 neurons, 532 in oligodendrocyte precursor cells, and 413 in astrocytes, against permutation null means near zero, and in HBCC 682 genes in layer 2/3 neurons and 674 in layer 2/3-6 neurons. The three interneuron subtypes therefore replicate strongly as part of broader age-associated expression change across cell types, and an exclusive interneuron-targeted claim is not supported. PsychAD adjudication of the replicated interneuron genes could not be run: PsychAD is absent from the Census release used here, and its pseudobulk resource requires Synapse authentication. The full replicated gene list (768 genes) is in the repository as results/reviewer-rev2-interneuron-genes-moderated.csv.

In the same interneurons, whole vector drift was null after depth adjustment in the original analysis. The pattern is therefore coordinated change in specific genes within specific cell types, not a general displacement of the whole expression profile. No protein, firing, or other functional phenotype was measured in these cells, so this result supports no functional claim and no cognition claim.

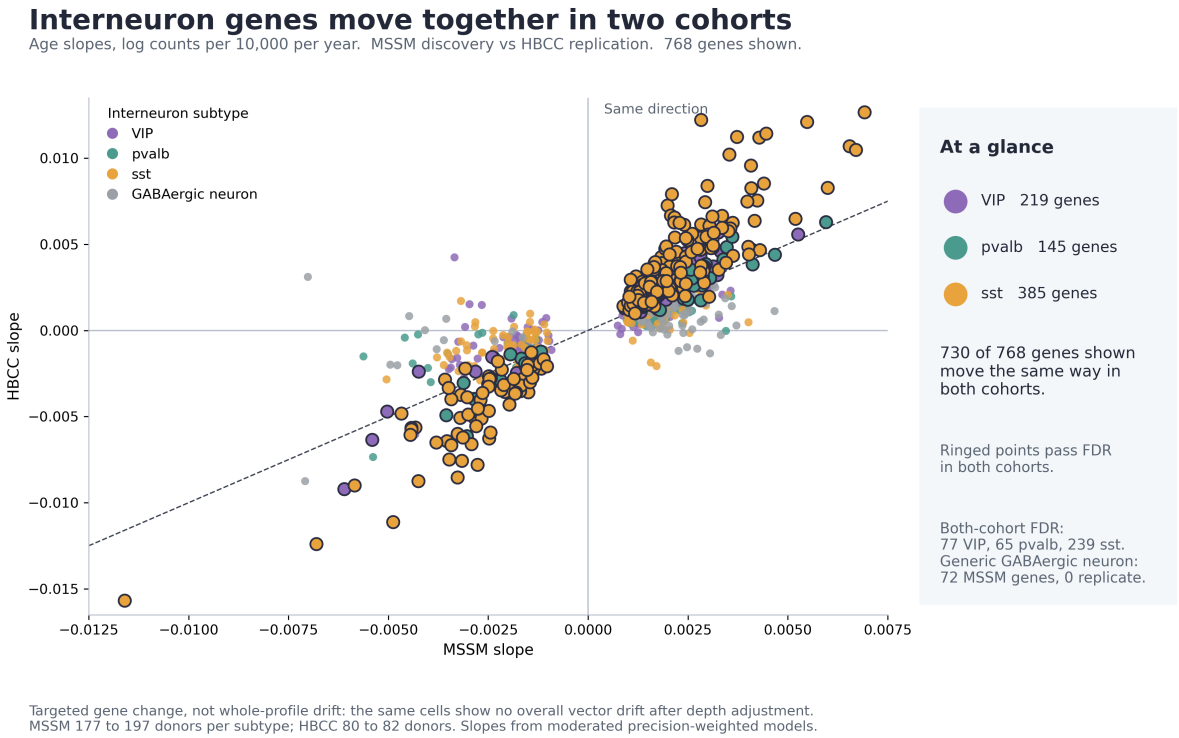


Figure 4: Figure 4. Interneuron gene changes. Each point is one gene in one interneuron subtype, with the generic GABAergic neuron type in grey; the axes are its MSSM age slope and its HBCC age slope, in log counts per 10,000 per year, from depth adjusted donor pseudobulk models with moderated standard errors. All 768 moderated MSSM false discovery rate genes are shown. Ringed points pass false discovery rate in both cohorts.

Off-program expression in mouse lung: no significant increase detected, with wide intervals

The direct test of leaky, irrelevant gene transcription was run in mouse lung only. Off program expression is the fraction of transcripts from genes that are uncommon in young cells of the target type but characteristic of another lung cell type. The original analysis measured this outcome only after downsampling every cell to 1,500 UMIs, which loses rare transcripts first. Depth sensitivity analyses therefore ran a depth sweep at 500, 1,500, and 3,000 UMIs per cell and at full depth, with young reference marker definitions fixed across depths, plus a library size covariate arm at full depth. The

1,500 UMI point estimates reproduce the original global effects. Effects in the table below are per 15 months.

Depth	Off-program effect	95 percent CI	p value	Entropy effect	95 percent CI	p value
500 UMIs	+0.2537	-0.2237 to +0.7311	0.2751	-0.0716	-0.4992 to +0.3559	0.7260
1,500 UMIs	+0.2438	-0.2349 to +0.7225	0.2948	-0.0789	-0.5135 to +0.3557	0.7041
3,000 UMIs	+0.1726	-0.3506 to +0.6958	0.4928	+0.0080	-0.4144 to +0.4304	0.9683
Full depth	+0.1743	-0.3557 to +0.7044	0.4940	-0.0906	-0.5517 to +0.3704	0.6812
Full depth, lib. size adj.	+0.1590	-0.3358 to +0.6539	0.5038	-0.0602	-0.3522 to +0.2318	0.6668

No statistically significant increase in off-program expression was detected in mouse lung at the tested depths, but confidence intervals were wide and biologically relevant increases were not excluded. No depth produced a significant positive global off program association. The full depth null does not prove biological stability: the off program minimum detectable effect at 80 percent power was 0.6191 standard deviations, and equivalence was established at neither the 0.1 nor the 0.25 standard deviation margin. The entropy null is in the same position, with a minimum detectable effect of 0.5621 standard deviations and no equivalence at either margin.

A separate direct endpoint tells a different story about mouse lung, and the two must not be merged. The cell type average expression vector analysis, rerun without downsampling, gave an unadjusted Hellinger global effect of plus 0.2914 standard deviations per 15 months (95 percent confidence interval minus 0.1203 to plus 0.7030, p = 0.1653). After mean log UMI adjustment it was plus 0.2626 standard deviations (95 percent confidence interval plus 0.0806 to plus 0.4445, p = 0.00468). This endpoint measures displacement from young profiles. It is not off program expression and it is not entropy.

Adult only models, nonlinear age, and annotation stability

Every public human vector and composition donor was already at least 20 years old. The minimum age across both analyses was 20, so adult only refitting uses the same donor sets and changes nothing for these public analyses. Developmental donors in PsychAD belong to the separate signature transport analysis. Nonlinear models used a fixed three degree of freedom cubic B spline. MSSM had two linear false discovery rate hits and no nonlinear hits in the vector analysis, and six linear hits and one nonlinear hit in the composition analysis. Integrated lung had two linear composition hits and no nonlinear hits. These results should not be summarized as confirmation of linear age effects, because nonlinearity and cohort specific age coverage can change individual conclusions.

Annotation stability limits how far the composition results can be pushed. No per cell annotation confidence probability was available in the mouse or public human inputs, and doublet scores were not relabeled as annotation confidence. Age related differences in excluded or unassigned cells were detected in integrated human lung (Spearman rho = +0.4334, q = 0.0251 across 48 ages) and in mouse pancreas (rho = +1.0 across four age groups, a result with only four groups that should not be generalized). A fixed canonical marker analysis found eight false discovery rate significant changes across 187 tests, seven in human and one in mouse, including increased microglial C1QA in MSSM (slope +0.00522 log counts per 10,000 per year, q = 0.000280) and decreased mouse cardiac fibroblast COL1A1 (slope -0.0397, q = 0.00617). These findings do not demonstrate identity loss, because marker expression can change without loss of cell identity. They do mean that identity loss cannot be excluded as an explanation for any affected composition result.

Exploratory and supplementary arms

The following arms are exploratory or supplementary under the working scope of this revision. They are reported so that the record is complete, and none of them drives the main claims.

In fruit fly, gene effect concordance was plus 0.1936 as a reference-only value, and the bidirectional vector hits were not independent at the gene level: among 703 pairs of the 38 depth adjusted vector

hit cell types, 537 pairs shared at least one false discovery rate significant gene and 480 pairs had hypergeometric q below 0.05, with a median Jaccard index of 0.0130. Gene Ontology enrichment was concentrated in specific units rather than one common pathway across all hits. The fly whole-vector effect is null: head minus 0.142 ($p = 0.60$) and body plus 0.047 ($p = 0.86$) at full depth. In worm, the vector result depends on the reference: with a day 1 reference, the depth-adjusted effect is plus 1.59 to plus 1.68 standard deviations (p less than 0.0001) across adjustment forms, while the balanced day 6 reference gave plus 0.5319 in the earlier analysis (parametric $p = 0.0189$, permutation $p = 0.0535$). Day 6 is the balanced reference because it includes both sequencing platforms, so the worm vector effect is reference sensitive and is not presented as a settled number. In zebrafish, the broad-compartment vector effect was plus 0.71 to plus 0.75 ($p = 0.004$ to 0.012), while the fine grained effect was plus 0.2211 with a wide interval (95 percent confidence interval minus 0.3588 to plus 0.8009) in six fish. In naked mole rat, the vector point estimate was minus 0.4744 (exact $p = 0.2$) in five animals with skin only data, and the concordance analysis was unresolved because the permutation null itself was already high (observed +0.6146 against a null mean of +0.6423, $p = 0.7000$). Neither arm supports a verdict on stability or on a difference from other species. Naked mole rats are long lived animals whose mortality does not rise with age [18], which is context for the null, not evidence for it.

Discussion

The central result is a constraint. At the transcriptomic level, aging in mouse and human is cell type specific: composition changes in specific cell types in specific organs, gene changes concentrated in specific cell types, an overall expression displacement that is robust and distributed in mouse and small and imprecise overall in human, and no consistent increase in off program expression at any measured depth. No uniform transcriptomic program was detected in the surveyed datasets. The concordance result adds the needed nuance: gene level age effects do share direction across cell types within mouse tissues, so the local pattern is coordinated, not random. Coordination of direction is not a uniform program of deterioration, and this paper does not claim that it is. Upstream causes are not settled by these data, and shared mechanisms with cell-type-specific consequences are not excluded.

A middle position remains fully compatible with the data, as a compatibility statement and not as a claim: a common maintenance vulnerability, shared across cells in how genome, epigenome, proteome, and energy are maintained over time, with divergent local manifestation that depends on each cell type and its chromatin landscape, repair capacity, division behaviour, niche, and exposure. On that view, a clock can track age across mammals while the consequences of age in any given cell type stay local. Tracking time is not the same as causing dysfunction, and a shared vulnerability is not the same as one uniform transcriptional program.

Which downstream predictions were tested, and what the data say

Three downstream predictions of a uniform program were tested. First, cells should lose identity everywhere and transcribe more genes that do not belong to their program. The off-program test, the most direct transcriptomic test of leaky transcription in this study, found no significant increase in mouse lung at any tested depth, but its interval is wide. No statistically significant increase in off-program expression was detected in mouse lung at the tested depths, but confidence intervals were wide and biologically relevant increases were not excluded. That prediction is therefore untested in effect: the data neither confirm it nor exclude it. Second, whole expression profiles should move away from young references in a shared direction everywhere. Mouse profiles do drift, and the drift is distributed across cell types and survives depth adjustment, but human profiles show only a small overall change whose interval cannot distinguish no effect from a mouse-sized effect, and the drift is concentrated in the young to mature transition rather than continuing past 18 months. That prediction holds in mouse

and is unresolved in human. Third, cell-type fractions should shift in conserved directions. Fractions shift only locally, and the clearest human lung examples collapse into between-study associations once study is modeled. That prediction fails as stated. The proposal that DNA damage, followed by repair that scrambles epigenetic information, drives downstream inflammation, fibrosis, and transposon derepression [13] is neither proved nor disproved here: its upstream steps were not measured. Chromatin was not measured. DNA damage was not measured. Transposon RNA was not quantified. Protein and cell function were not measured. Actual turnover was not measured. What was measured is the downstream transcriptomic pattern such a theory would have to produce, and that pattern is local rather than uniform. Reprogramming work showing recovery of youthful epigenetic information in specific settings [14] is consistent with epigenetic information mattering for aging, and it does not show that a uniform transcriptional program exists. Fibrosis, in particular, is not demonstrated by any result here: fraction changes are changes in shares of recovered cells, not measurements of extracellular matrix deposition or of tissue takeover.

The next step that follows from these results is a single decisive measurement: same-cell multiome data, measuring chromatin and RNA in the same cells, in the cell types that do and do not change. If a shared upstream process exists, its chromatin footprint should appear in the same cells whose RNA changes, and it should be absent or different where RNA does not change. Until that measurement exists, the constraint stands as stated and no further verdict on whole theories is offered.

For measurement practice, the results argue for cell type resolved designs as the default, with depth modelled in every distance or detection analysis, with composition and within cell expression reported as separate findings, and with multiplicity handled across the full family of tests actually run. The concrete starting point offered here for directed validation is the replicated interneuron gene changes, on evidential strength. The mouse-derived signature is not offered as a validation target: it did not transport beyond matched random signatures in either cohort.

Limitations

These limitations are part of the results, not qualifications added after them.

The study is cross sectional and observational throughout. There is no perturbation, no longitudinal follow up, and no causal identification. Age associations can reflect cohort composition, technical variation, development, disease, or postmortem effects.

No equivalence claim is made in this paper. The human overall vector interval cannot distinguish no effect from a mouse-sized effect, and it is reported as imprecise rather than as evidence of absence. The off program, entropy, naked mole rat, and zebrafish nulls are likewise imprecise rather than evidence of stability.

Fly and worm samples are pools, not individually tracked animals, so within pool variation is not available to the inference. The naked mole rat arm has five animals, with skin only and group level ages, and the zebrafish fine grained arm is small. Neither can support claims of stability, protection, or species difference.

PsychAD and the public MSSM and HBCC cohorts contain exact identifier confirmed overlapping donors, so PsychAD analyses are not independent replication of the public cohorts. The MSSM to HBCC interneuron comparison uses different source banks.

Annotation is fixed while marker expression changes with age, and excluded cell fractions differ by age in at least two datasets. Identity loss therefore cannot be excluded as an explanation for any affected composition result.

Cell turnover was not measured. Published estimates of turnover vary widely by cell type [17], so no result here can be attributed to division history or to turnover, and no turnover claim is made.

Depth, sequencing platform, and reference choice affect the measurements throughout. Platform and age are partly entangled in worm, and the worm vector effect changes size and security with the

reference age.

Cross species effect sizes are standardized within each dataset. They are not directly comparable as measures of biological importance, because tissues, age windows, assays, annotation resolution, and lifespan coverage differ between species.

Conclusions

No uniform transcriptomic program was detected in the surveyed datasets. Fractions shift only in specific cell types in specific organs, whole expression profiles drift in mouse but show only a small and imprecise overall change in human, and the durable human finding is gene change in three interneuron subtypes that replicates across two cohorts, as part of broader age-associated expression change across cell types. The mouse-derived signature did not transport to either human cohort beyond expression-matched random signatures. No statistically significant increase in off-program expression was detected in mouse lung at the tested depths, but confidence intervals were wide and biologically relevant increases were not excluded. Gene level effects share direction across cell types more than a donor-level null allows in mouse, which shows coordination without a uniform breakdown. Upstream causes are not settled, and shared mechanisms with cell-type-specific consequences are not excluded. Any shared upstream process must therefore have consequences that differ by cell type, and the decisive next step is a same-cell multiome measurement in the cell types that do and do not change.

Data Availability

The analysis code and aggregate result tables are in the Agescape repository at https://github.com/Abolfazl-Alipour/agescape. Dataset accessions are given in the Materials and Methods. A Zenodo archive of the analysis code and result tables is pending; no DOI is claimed. The results published here are in whole or in part based on data obtained from the AD Knowledge Portal (NIH-NIA, U24AG061340). Controlled access PsychAD donor level values (Synapse syn52396927, pseudobulk resource syn62064518) are available from the source resource under its own terms and were not redistributed; only aggregate results are shared here.

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Competing Interests

The author declares no conflict of interest. [Pending author confirmation.]

Ethics Statement

This study performed no new human or animal work. All data analyzed are from published public cohorts and controlled access resources used under their own terms, as described in the Materials and Methods. No new human participants or animals were recruited, no interventions were performed, and no identifiable individual level data are reported. [Draft statement, pending author confirmation. No ethics approval number is claimed.]

Supporting Information

No supporting figures are included in this revision. The following text note supports the main text.

Meta-analysis note. A random effects meta-analysis of the global Hellinger effects across all six species, using the balanced worm day 6 reference, gave a pooled effect of plus 0.0721 standard deviations per species specific age scale, with a 95 percent confidence interval of minus 0.0232 to plus 0.1674 (p = 0.1383) and substantial heterogeneity (Q = 15.20 on 5 degrees of freedom, p = 0.00956). With the worm day 1 reference the pooled effect was plus 0.1454 (95 percent confidence interval minus 0.0182 to plus 0.3090, p = 0.0815). The pooled numbers are unstable to the worm reference choice, and heterogeneity Q is not evidence against universality: it measures variation in effect size across species, which a uniform program with species-specific magnitude would also produce. The meta-analysis is therefore reported here only as a sensitivity note, and it is not used to support any main-text claim.

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